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TestAmerica

ANALYTICAL REPORT

Eurofins TestAmerica, Michigan
10448 Citation Drive
Suite 200
Brighton, MI 48116
Tel: (810)229-2763

Laboratory Job ID: 190-19915-1
Client Project/Site: City of Cadillac WWTP PFAS

For:
City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
6/25/2019 3:57:38 PM

Sue Schafer, Project Manager II
(810)229-2763
sue.schafer@testamericainc.com

LINKS

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This report has been electronically signed and authorized by the signatory. Electronic signature is intended to be the legally binding equivalent of a traditionally handwritten signature.

Results relate only to the items tested and the sample(s) as received by the laboratory.

Table of Contents

Sample Summary

Lab Sample ID	Client Sample ID	Matrix	Collected	Received	Asset ID
190-19915-1	Effluent	Water	06/04/19 13:15	06/05/19 12:15	
190-19915-2	Effluent Duplicate	Water	06/04/19 13:17	06/05/19 12:15	
190-19915-3	Equipment Blank	Water	06/04/19 13:12	06/05/19 12:15	
190-19915-4	Field Blank	Water	06/04/19 13:08	06/05/19 12:15	

Job ID: 190-19915-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative
190-19915-1

Comments
No additional comments.

Receipt
The samples were received on 6/5/2019 12:15 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 3.9º C.

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.

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Sample Administration Receipt Documentation Log

Doc Log ID: 250845

Group Number(s): 2047572

Client: Eurofins TestAmerica

Delivery and Receipt Information

Delivery Method:	Fed Ex	Arrival Timestamp:	06/06/2019 8:00
Number of Packages:	1	Number of Projects:	1
State/Province of Origin:	MI		

Arrival Condition Summary

Shipping Container Sealed:	Yes	Sample IDs on COC match Containers:	Yes
Custody Seal Present:	Yes	Sample Date/Times match COC:	Yes
Custody Seal Intact:	Yes	VOA Vial Headspace ≥ 6mm:	N/A
Samples Chilled:	Yes	Total Trip Blank Qty:	2
Paperwork Enclosed:	Yes	Trip Blank Type:	see COC
Samples Intact:	Yes	Air Quality Samples Present:	No
Missing Samples:	No		
Extra Samples:	No		
Discrepancy in Container Qty on COC:	No		

Unpacked by Cory Jeremiah (10469) at 17:27 on 06/06/2019

Samples Chilled Details

Thermometer Types: DT = Digital (Temp. Bottle) IR = Infrared (Surface Temp) All Temperatures in °C.

Cooler #	Thermometer ID	Corrected Temp	Therm. Type	Ice Type	Ice Present?	Ice Container	Elevated Temp?
1	32170023	3.2	IR	Wet	Y	Loose	N

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Lancaster, PA 17605-2425
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ANALYSIS REPORT

Prepared by:

Eurofins Lancaster Laboratories Environmental
2425 New Holland Pike
Lancaster, PA 17601

Prepared for:

TestAmerica Brighton MI
10448 Citation Drive
Suite 200
Brighton MI 48116

Report Date: June 21, 2019 13:39

Project: City of Cadillac WWTP PFAS

Account #: 01042
Group Number: 2047572
Release Number: 190-19915-1
State of Sample Origin: MI

Electronic Copy To TestAmerica Attn: Sue Schafer

Respectfully Submitted,

Wendy A. Kozma

Wendy A. Kozma
Principal Specialist Group Leader

(717) 556-7257

To view our laboratory's current scopes of accreditation please go to https://www.eurofinsus.com/environment-testing/laboratories/eurofins-lancaster-laboratories-environmental/certifications-and-accreditations-eurofins-lancaster-laboratories-environmental/ . Historical copies may be requested through your project manager.

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SAMPLE INFORMATION

Client Sample Description	Sample Collection Date/Time	ELLE#
Effluent (190-19915-1) Water	06/04/2019 13:15	1074244
Effluent Duplicate (190-19915-2) Water	06/04/2019 13:17	1074245
Equipment Blank (190-19915-3) Water	06/04/2019 13:12	1074246
Field Blank (190-19915-4) Water	06/04/2019 13:08	1074247

The specific methodologies used in obtaining the enclosed analytical results are indicated on the Laboratory Sample Analysis Record.

1 2 3 4 5 6 7 8

Sample Description: Effluent (190-19915-1) Water
City of Cadillac WWTP

Project Name: City of Cadillac WWTP PFAS

Submittal Date/Time: 06/06/2019 08:00
Collection Date/Time: 06/04/2019 13:15

TestAmerica Brighton MI
ELLE Sample #: WW 1074244
ELLE Group #: 2047572
Matrix: Water

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		ng/l	ng/l	ng/l	
14473	4:2-Fluorotelomersulfonic acid	757124-72-4	N.D.	0.82	2.5	1
14473	6:2-Fluorotelomersulfonic acid	27619-97-2	2.4	0.82	1.6	1
14473	8:2-Fluorotelomersulfonic acid	39108-34-4	N.D.	1.6	4.9	1
14473	NEtFOSAA	2991-50-6	1.1 J	0.82	2.5	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	4.4	0.82	2.5	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorobutanesulfonic acid	375-73-5	49	0.25	0.82	1
14473	Perfluorodecanesulfonic acid	335-77-3	N.D.	0.49	1.6	1
14473	Perfluoroheptanesulfonic acid	375-92-8	0.40 J	0.33	1.6	1
14473	Perfluorohexanesulfonic acid	355-46-4	16	0.33	1.6	1
14473	Perfluorononanesulfonic acid	68259-12-1	N.D.	0.49	1.6	1
14473	Perfluorooctanesulfonic acid	1763-23-1	7.8	0.33	1.6	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	22	1.6	4.9	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	1.5 J	0.74	1.6	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.41	1.6	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	6.7	0.33	0.82	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	55	0.33	1.6	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	0.98 J	0.33	1.6	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	16	0.25	0.82	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.41	2.5	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	18	1.6	4.9	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	7.3	0.33	1.6	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.25	0.82	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.33	0.82	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.33	1.6	1

The recovery of the extraction standard d5-NEtFOSAA was outside QC acceptance criteria in the opening CCV associated with this sample. Since the recovery of the d5-NEtFOSAA was within acceptance criteria in the sample, the data is reported.

The recoveries for several labeled compounds used as extraction standards were outside of QC acceptance limits as noted on the QC Summary due to the matrix of the sample.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	24 PFAS - MI List	EPA 537 Version 1.1 Modified	1	19165016	06/18/2019 21:22	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	2	19165016	06/14/2019 16:00	Anthony C Polaski	1

*=This limit was used in the evaluation of the final result

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Analysis Report

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Sample Description:	**Effluent Duplicate (190-19915-2) Water** **City of Cadillac WWTP**	**TestAmerica Brighton MI**	
		ELLE Sample #:	**WW 1074245**
		ELLE Group #:	**2047572**
Project Name:	**City of Cadillac WWTP PFAS**	**Matrix: Water**	
Submittal Date/Time:	06/06/2019 08:00		
Collection Date/Time:	06/04/2019 13:17		

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		ng/l	ng/l	ng/l	
14473	4:2-Fluorotelomersulfonic acid	757124-72-4	N.D.	0.80	2.4	1
14473	6:2-Fluorotelomersulfonic acid	27619-97-2	2.2	0.80	1.6	1
14473	8:2-Fluorotelomersulfonic acid	39108-34-4	N.D.	1.6	4.8	1
14473	NEtFOSAA	2991-50-6	1.2 J	0.80	2.4	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	3.8	0.80	2.4	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorobutanesulfonic acid	375-73-5	49	0.24	0.80	1
14473	Perfluorodecanesulfonic acid	335-77-3	N.D.	0.48	1.6	1
14473	Perfluoroheptanesulfonic acid	375-92-8	0.35 J	0.32	1.6	1
14473	Perfluorohexanesulfonic acid	355-46-4	17	0.32	1.6	1
14473	Perfluorononanesulfonic acid	68259-12-1	N.D.	0.48	1.6	1
14473	Perfluorooctanesulfonic acid	1763-23-1	7.1	0.32	1.6	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	23	1.6	4.8	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	1.3 J	0.72	1.6	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.40	1.6	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	7.4	0.32	0.80	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	59	0.32	1.6	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	0.99 J	0.32	1.6	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	15	0.24	0.80	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.40	2.4	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	17	1.6	4.8	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	8.7	0.32	1.6	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.24	0.80	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.32	0.80	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.32	1.6	1

The recovery of the extraction standard d5-NEtFOSAA was outside QC acceptance criteria in the opening CCV associated with this sample. Since the recovery of the d5-NEtFOSAA was within acceptance criteria in the sample, the data is reported.

The recoveries for several labeled compounds used as extraction standards were outside of QC acceptance limits as noted on the QC Summary due to the matrix of the sample.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	24 PFAS - MI List	EPA 537 Version 1.1 Modified	1	19165016	06/18/2019 21:31	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	2	19165016	06/14/2019 16:00	Anthony C Polaski	1

*=This limit was used in the evaluation of the final result

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Analysis Report

Sample Description:	**Equipment Blank (190-19915-3) Water City of Cadillac WWTP**	**TestAmerica Brighton MI**	
		ELLE Sample #:	**WW 1074246**
		ELLE Group #:	**2047572**
Project Name:	**City of Cadillac WWTP PFAS**	**Matrix: Water**	
Submittal Date/Time:	06/06/2019 08:00		
Collection Date/Time:	06/04/2019 13:12		

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		**ng/l**	**ng/l**	**ng/l**	
14473	4:2-Fluorotelomersulfonic acid	757124-72-4	N.D.	0.88	2.6	1
14473	6:2-Fluorotelomersulfonic acid	27619-97-2	N.D.	0.88	1.8	1
14473	8:2-Fluorotelomersulfonic acid	39108-34-4	N.D.	1.8	5.3	1
14473	NEtFOSAA	2991-50-6	N.D.	0.88	2.6	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	N.D.	0.88	2.6	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorobutanesulfonic acid	375-73-5	N.D.	0.26	0.88	1
14473	Perfluorodecanesulfonic acid	335-77-3	N.D.	0.53	1.8	1
14473	Perfluoroheptanesulfonic acid	375-92-8	N.D.	0.35	1.8	1
14473	Perfluorohexanesulfonic acid	355-46-4	N.D.	0.35	1.8	1
14473	Perfluorononanesulfonic acid	68259-12-1	N.D.	0.53	1.8	1
14473	Perfluorooctanesulfonic acid	1763-23-1	N.D.	0.35	1.8	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.8	5.3	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.79	1.8	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.44	1.8	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.35	0.88	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.35	1.8	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.35	1.8	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.26	0.88	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.44	2.6	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.8	5.3	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.35	1.8	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.26	0.88	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.35	0.88	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.35	1.8	1

The recovery of the extraction standard d5-NEtFOSAA was outside QC acceptance criteria in the opening CCV associated with this sample. Since the recovery of the d5-NEtFOSAA was within acceptance criteria in the sample, the data is reported.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	24 PFAS - MI List	EPA 537 Version 1.1 Modified	1	19165016	06/18/2019 21:40	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	2	19165016	06/14/2019 16:00	Anthony C Polaski	1

*=This limit was used in the evaluation of the final result

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Sample Description: Field Blank (190-19915-4) Water
City of Cadillac WWTP

Project Name: City of Cadillac WWTP PFAS

TestAmerica Brighton MI
ELLE Sample #: WW 1074247
ELLE Group #: 2047572
Matrix: Water

Submittal Date/Time: 06/06/2019 08:00
Collection Date/Time: 06/04/2019 13:08

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		ng/l	ng/l	ng/l	
14473	4:2-Fluorotelomersulfonic acid	757124-72-4	N.D.	0.84	2.5	1
14473	6:2-Fluorotelomersulfonic acid	27619-97-2	N.D.	0.84	1.7	1
14473	8:2-Fluorotelomersulfonic acid	39108-34-4	N.D.	1.7	5.0	1
14473	NEtFOSAA	2991-50-6	N.D.	0.84	2.5	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	N.D.	0.84	2.5	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorobutanesulfonic acid	375-73-5	N.D.	0.25	0.84	1
14473	Perfluorodecanesulfonic acid	335-77-3	N.D.	0.50	1.7	1
14473	Perfluoroheptanesulfonic acid	375-92-8	N.D.	0.33	1.7	1
14473	Perfluorohexanesulfonic acid	355-46-4	N.D.	0.33	1.7	1
14473	Perfluorononanesulfonic acid	68259-12-1	N.D.	0.50	1.7	1
14473	Perfluorooctanesulfonic acid	1763-23-1	N.D.	0.33	1.7	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.7	5.0	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.75	1.7	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.42	1.7	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.33	0.84	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.33	1.7	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.33	1.7	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.25	0.84	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.42	2.5	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.7	5.0	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.33	1.7	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.25	0.84	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.33	0.84	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.33	1.7	1

The recovery of the extraction standard d5-NEtFOSAA was outside QC acceptance criteria in the opening CCV associated with this sample. Since the recovery of the d5-NEtFOSAA was within acceptance criteria in the sample, the data is reported.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	24 PFAS - MI List	EPA 537 Version 1.1 Modified	1	19165016	06/18/2019 21:49	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	2	19165016	06/14/2019 16:00	Anthony C Polaski	1

*=This limit was used in the evaluation of the final result

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Quality Control Summary

Client Name: TestAmerica Brighton MI   Group Number: 2047572
Reported: 06/21/2019 13:39

Matrix QC may not be reported if insufficient sample or site-specific QC samples were not submitted. In these situations, to demonstrate precision and accuracy at a batch level, a LCS/LCSD was performed, unless otherwise specified in the method.

All Inorganic Initial Calibration and Continuing Calibration Blanks met acceptable method criteria unless otherwise noted on the Analysis Report.

Method Blank

Analysis Name	Result ng/l	MDL** ng/l	LOQ ng/l
Batch number: 19165016	Sample number(s): 1074244-1074247		
4:2-Fluorotelomersulfonic acid	N.D.	1.0	3.0
6:2-Fluorotelomersulfonic acid	N.D.	1.0	2.0
8:2-Fluorotelomersulfonic acid	N.D.	2.0	6.0
NEtFOSAA	N.D.	1.0	3.0
NMeFOSAA	N.D.	1.0	3.0
Perfluorobutanesulfonic acid	N.D.	0.30	1.0
Perfluorodecanesulfonic acid	N.D.	0.60	2.0
Perfluoroheptanesulfonic acid	N.D.	0.40	2.0
Perfluorohexanesulfonic acid	N.D.	0.40	2.0
Perfluorononanesulfonic acid	N.D.	0.60	2.0
Perfluorooctanesulfonic acid	N.D.	0.40	2.0
Pfba-Perfluorobutanoic Acid	N.D.	2.0	6.0
Pfda-Perfluorodecanoic Acid	N.D.	0.90	2.0
Pfdoda-Perfluorododecanoic	N.D.	0.50	2.0
Pfhpa-Perfluoroheptanoic Acid	N.D.	0.40	1.0
Pfhxa-Perfluorohexanoic Acid	N.D.	0.40	2.0
Pfna-Perfluorononanoic Acid	N.D.	0.40	2.0
Pfoa-Perfluorooctanoic Acid	N.D.	0.30	1.0
Pfosa-Perfluorooctanesulfonami	N.D.	0.50	3.0
Pfpea-Perfluoropentanoic Acid	N.D.	2.0	6.0
Pfpes-Perfluoropentanesulfonat	N.D.	0.40	2.0
Pfteda-Perfluorotetradecanoic	N.D.	0.30	1.0
Pftrda-Perfluorotridecanoic Ac	N.D.	0.40	1.0
Pfunda-Perfluoroundecanoic Aci	N.D.	0.40	2.0

LCS/LCSD

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Batch number: 19165016	Sample number(s): 1074244-1074247								
4:2-Fluorotelomersulfonic acid	14.94	14.19	14.94	14.63	95	98	82-152	3	30
6:2-Fluorotelomersulfonic acid	15.17	13.89	15.17	15.67	92	103	66-155	12	30
8:2-Fluorotelomersulfonic acid	15.33	14.37	15.33	12.48	94	81	66-148	14	30
NEtFOSAA	5.44	5.65	5.44	5.97	104	110	55-169	5	30
NMeFOSAA	5.44	5.97	5.44	5.69	110	105	44-147	5	30

*- Outside of specification
**-This limit was used in the evaluation of the final result for the blank
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

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Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 06/21/2019 13:39

Group Number: 2047572

LCS/LCSD (continued)

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Perfluorobutanesulfonic acid	4.81	4.63	4.81	4.88	96	101	73-128	5	30
Perfluorodecanesulfonic acid	5.24	5.33	5.24	5.00	102	96	60-135	6	30
Perfluoroheptanesulfonic acid	5.18	5.15	5.18	5.21	99	101	64-135	1	30
Perfluorohexanesulfonic acid	5.14	4.87	5.14	5.14	95	100	71-131	5	30
Perfluorononanesulfonic acid	5.22	5.02	5.22	5.34	96	102	66-133	6	30
Perfluorooctanesulfonic acid	5.20	4.75	5.20	4.82	91	93	67-138	1	30
Pfba-Perfluorobutanoic Acid	5.44	6.02	5.44	6.53	111	120	74-142	8	30
Pfda-Perfluorodecanoic Acid	5.44	6.00	5.44	5.42	110	100	69-148	10	30
Pfdoda-Perfluorododecanoic	5.44	5.78	5.44	6.07	106	112	75-136	5	30
Pfhpa-Perfluoroheptanoic Acid	5.44	5.81	5.44	5.60	107	103	76-140	4	30
Pfhxa-Perfluorohexanoic Acid	5.44	5.34	5.44	5.65	98	104	75-135	6	30
Pfna-Perfluorononanoic Acid	5.44	5.45	5.44	5.71	100	105	72-148	5	30
Pfoa-Perfluorooctanoic Acid	5.44	5.79	5.44	5.35	106	98	72-138	8	30
Pfosa-Perfluorooctanesulfonami	5.44	5.37	5.44	5.72	99	105	65-164	6	30
Pfpea-Perfluoropentanoic Acid	5.44	5.21	5.44	5.38	96	99	74-134	3	30
Pfpes-Perfluoropentanesulfonat	5.10	4.96	5.10	5.12	97	100	76-127	3	30
Pfteda-Perfluorotetradecanoic	5.44	5.37	5.44	6.07	99	112	74-135	12	30
Pftrda-Perfluorotridecanoic Ac	5.44	4.77	5.44	5.44	88	100	61-145	13	30
Pfunda-Perfluoroundecanoic Aci	5.44	6.11	5.44	5.13	112	94	75-146	17	30

Labeled Isotope Quality Control

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: 24 PFAS - MI List
Batch number: 19165016

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
1074244	67	69	101	193*	46	66
1074245	71	68	104	216*	48	72
1074246	64	65	62	65	58	62
1074247	75	76	73	74	72	73
Blank	71	73	71	62	71	74
LCS	85	89	83	79	87	90
LCSD	74	76	73	64	67	74
Limits:	33-123	31-157	26-148	21-182	35-138	34-126

	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
1074244	65	198*	69	69	78	67
1074245	66	223*	79	77	83	73
1074246	62	67	64	63	61	67

*- Outside of specification
**-This limit was used in the evaluation of the final result for the blank
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

2425 New Holland Pike, Lancaster, PA 17601 • 717-656-2300 • Fax: 717-656-6766 • www.EurofinsUS.com/LancLabsEnv

Quality Control Summary

Client Name: TestAmerica Brighton MI   Group Number: 2047572
Reported: 06/21/2019 13:39

Labeled Isotope Quality Control (continued)

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: 24 PFAS - MI List
Batch number: 19165016

	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
1074247	71	71	73	74	70	74
Blank	76	70	71	76	73	69
LCS	88	90	86	82	83	73
LCSD	73	69	76	74	69	78
Limits:	35-126	32-170	48-122	50-121	41-144	47-125

	13C2-8:2-FTS	d3-NMeFOSAA	13C7-PFUnDA	d5-NEtFOSAA	13C2-PFDoDA	13C2-PFTeDA
1074244	112	76	67	80	57	27
1074245	128	81	76	92	64	41
1074246	64	70	68	72	61	61
1074247	77	88	79	84	83	70
Blank	70	75	70	81	64	56
LCS	74	81	75	90	75	74
LCSD	86	84	83	75	68	64
Limits:	27-164	30-127	30-128	30-142	39-130	26-119

	13C8-PFOSA
1074244	44
1074245	45
1074246	63
1074247	82
Blank	74
LCS	87
LCSD	79
Limits:	11-127

*- Outside of specification
**-This limit was used in the evaluation of the final result for the blank
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

Eurofins TestAmerica, Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone: 810-229-2763 Fax: 810-229-0000

Chain of Custody Record

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A-1042 G-2047572 S-1074244-47

Client Information (Sub Contract Lab)	Sampler:	Lab PM: Schafer, Sue	Carrier Tracking No(s):	COC No: 190-23253.1
Client Contact: Shipping/Receiving	Phone:	E-Mail: sue.schafer@testamericainc.com	State of Origin: Michigan	Page: Page 1 of 1
Company: Eurofins Lancaster Laboratories Env LLC		Accreditations Required (See note):		Job #: 190-19915-1
Address: 2425 New Holland Pike, ,	Due Date Requested: 6/21/2019			
City: Lancaster	TAT Requested (days):			
State, Zip: PA, 17601				
Phone: 717-656-2300(Tel)	PO #:			
Email:	WO #:			
Project Name: City of Cadillac WWTP PFAS	Project #: 19001884			
Site:	SSOW#:			

Analysis Requested

Preservation Codes:
A - HCL, B - NaOH, C - Zn Acetate, D - Nitric Acid, E - NaHSO4, F - MeOH, G - Amchlor, H - Ascorbic Acid, I - Ice, J - DI Water, K - EDTA, L - EDA
M - Hexane, N - None, O - AsNaO2, P - Na2O4S, Q - Na2SO3, R - Na2S2O3, S - H2SO4, T - TSP Dodecahydrate, U - Acetone, V - MCAA, W - pH 4-5, Z - other (specify)
Other:

Sample Identification - Client ID (Lab ID)	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	SUB (PFAS)/ PFAS	Total Number of containers	Special Instructions/Note:
			Preservation Code:						
Effluent (190-19915-1)	6/4/19	13:15 Eastern		Water			X	2	METHOD 537M - 24 PFAS COMPOUNDS
Effluent Duplicate (190-19915-2)	6/4/19	13:17 Eastern		Water			X	2	METHOD 537M - 24 PFAS COMPOUNDS
Equipment Blank (190-19915-3)	6/4/19	13:12 Eastern		Water			X	2	METHOD 537M - 24 PFAS COMPOUNDS
Field Blank (190-19915-4)	6/4/19	13:08 Eastern		Water			X	2	METHOD 537M - 24 PFAS COMPOUNDS

Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said compliance to TestAmerica Laboratories, Inc.

Possible Hazard Identification
Unconfirmed

Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)
☐ *Return To Client* ☐ *Disposal By Lab* ☐ *Archive For ______ Months*

Deliverable Requested: I, II, III, IV, Other (specify) Primary Deliverable Rank: 2

Special Instructions/QC Requirements:

Empty Kit Relinquished by:	Date:	Time:	Method of Shipment:

Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company
[signature]	4/5/19 1420	ETA			
			[signature]	6-6-19 800	ELLE

Custody Seals Intact: Δ Yes Δ No	Custody Seal No.:	Cooler Temperature(s) °C and Other Remarks: 3.2 °C

Ver: 01/16/2019

eurofins Lancaster Laboratories Environmental

Sample Administration Receipt Documentation Log

Doc Log ID: 250845

Group Number(s): 2047572

Client: Eurofins TestAmerica

Delivery and Receipt Information

Delivery Method:	Fed Ex	Arrival Timestamp:	06/06/2019 8:00
Number of Packages:	1	Number of Projects:	1
State/Province of Origin:	MI		

Arrival Condition Summary

Shipping Container Sealed:	Yes	Sample IDs on COC match Containers:	Yes
Custody Seal Present:	Yes	Sample Date/Times match COC:	Yes
Custody Seal Intact:	Yes	VOA Vial Headspace ≥ 6mm:	N/A
Samples Chilled:	Yes	Total Trip Blank Qty:	2
Paperwork Enclosed:	Yes	Trip Blank Type:	see COC
Samples Intact:	Yes	Air Quality Samples Present:	No
Missing Samples:	No		
Extra Samples:	No		
Discrepancy in Container Qty on COC:	No		

Unpacked by Cory Jeremiah (10469) at 17:27 on 06/06/2019

Samples Chilled Details

Thermometer Types: *DT = Digital (Temp. Bottle)* *IR = Infrared (Surface Temp)* *All Temperatures in °C.*

Cooler #	Thermometer ID	Corrected Temp	Therm. Type	Ice Type	Ice Present?	Ice Container	Elevated Temp?
1	32170023	3.2	IR	Wet	Y	Loose	N

2425 New Holland Pike
Lancaster, PA 17605-2425

T 717-656-2300
F 717-656-2681
www.LancasterLabs.com

Explanation of Symbols and Abbreviations

The following defines common symbols and abbreviations used in reporting technical data:

BMQL	Below Minimum Quantitation Level
C	degrees Celsius
cfu	colony forming units
CP Units	cobalt-chloroplatinate units
F	degrees Fahrenheit
g	gram(s)
IU	International Units
kg	kilogram(s)
L	liter(s)
lb.	pound(s)
m3	cubic meter(s)
meq	milliequivalents
mg	milligram(s)
mL	milliliter(s)
MPN	Most Probable Number
N.D.	non-detect
ng	nanogram(s)
NTU	nephelometric turbidity units
pg/L	picogram/liter
RL	Reporting Limit
TNTC	Too Numerous To Count
µg	microgram(s)
µL	microliter(s)
umhos/cm	micromhos/cm
MCL	Maximum Contamination Limit
<	less than
>	greater than
ppm	parts per million - One ppm is equivalent to one milligram per kilogram (mg/kg) or one gram per million grams. For aqueous liquids, ppm is usually taken to be equivalent to milligrams per liter (mg/l), because one liter of water has a weight very close to a kilogram. For gases or vapors, one ppm is equivalent to one microliter per liter of gas.
ppb	parts per billion
Dry weight basis	Results printed under this heading have been adjusted for moisture content. This increases the analyte weight concentration to approximate the value present in a similar sample without moisture. All other results are reported on an as-received basis.

Analytical test results meet all requirements of the associated regulatory program (i.e., NELAC (TNI), DoD, and ISO 17025) unless otherwise noted under the individual analysis.

Measurement uncertainty values, as applicable, are available upon request.

Tests results relate only to the sample tested. Clients should be aware that a critical step in a chemical or microbiological analysis is the collection of the sample. Unless the sample analyzed is truly representative of the bulk of material involved, the test results will be meaningless. If you have questions regarding the proper techniques of collecting samples, please contact us. We cannot be held responsible for sample integrity, however, unless sampling has been performed by a member of our staff.

This report shall not be reproduced except in full, without the written approval of the laboratory.

Times are local to the area of activity. Parameters listed in the 40 CFR Part 136 Table II as “analyze immediately” are not performed within 15 minutes.

WARRANTY AND LIMITS OF LIABILITY - In accepting analytical work, we warrant the accuracy of test results for the sample as submitted. THE FOREGOING EXPRESS WARRANTY IS EXCLUSIVE AND IS GIVEN IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED. WE DISCLAIM ANY OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING A WARRANTY OF FITNESS FOR PARTICULAR PURPOSE AND WARRANTY OF MERCHANTABILITY. IN NO EVENT SHALL EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL, LLC BE LIABLE FOR INDIRECT, SPECIAL, CONSEQUENTIAL, OR INCIDENTAL DAMAGES INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOSS OF PROFIT OR GOODWILL REGARDLESS OF (A) THE NEGLIGENCE (EITHER SOLE OR CONCURRENT) OF EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL AND (B) WHETHER EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL HAS BEEN INFORMED OF THE POSSIBILITY OF SUCH DAMAGES. We accept no legal responsibility for the purposes for which the client uses the test results. No purchase order or other order for work shall be accepted by Eurofins Lancaster Laboratories Environmental which includes any conditions that vary from the Standard Terms and Conditions, and Eurofins Lancaster Laboratories Environmental hereby objects to any conflicting terms contained in any acceptance or order submitted by client.

Data Qualifiers

Qualifier	Definition
C	Result confirmed by reanalysis
D1	Indicates for dual column analyses that the result is reported from column 1
D2	Indicates for dual column analyses that the result is reported from column 2
E	Concentration exceeds the calibration range
K1	Initial Calibration Blank is above the QC limit and the sample result is ND
K2	Continuing Calibration Blank is above the QC limit and the sample result is ND
K3	Initial Calibration Verification is above the QC limit and the sample result is ND
K4	Continuing Calibration Verification is above the QC limit and the sample result is ND
J (or G, I, X)	Estimated value >= the Method Detection Limit (MDL or DL) and < the Limit of Quantitation (LOQ or RL)
P	Concentration difference between the primary and confirmation column >40%. The lower result is reported.
P^	Concentration difference between the primary and confirmation column > 40%. The higher result is reported.
U	Analyte was not detected at the value indicated
V	Concentration difference between the primary and confirmation column >100%. The reporting limit is raised due to this disparity and evident interference.
W	The dissolved oxygen uptake for the unseeded blank is greater than 0.20 mg/L.
Z	Laboratory Defined - see analysis report

Additional Organic and Inorganic CLP qualifiers may be used with Form 1 reports as defined by the CLP methods.
Qualifiers specific to Dioxin/Furans and PCB Congeners are detailed on the individual Analysis Report.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
¤	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

Accreditation/Certification Summary

Laboratory: Eurofins TestAmerica, Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Michigan	State		0057	05-05-20
Michigan	State Program	5	57	05-05-20

Eurofins TestAmerica, Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone: 810-229-2763 Fax: 810-229-0000

Chain of Custody Record

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Client Information			
Client Contact: Cindy Tomaszewski	Sampler: CA Tomaszewski	Lab PM: Schafer, Sue	Carrier Tracking No(s):
Company: City of Cadillac Utilities	Phone: 231-775-2368	E-Mail: sue.schafer@testamericainc.com	COC No: 190-24202-1153.1
Address: 1121 Plett Road	Due Date Requested:		Page: Page 1 of 1
City: Cadillac	TAT Requested (days):		Job #:
State, Zip: MI, 49601			
Phone: 231-775-2368	PO #: 2019-~~018~~ TBD		
Email: lab@cadillac-mi.net	WO #:		
Project Name: City of Cadillac	Project #: 19001884		
Site: WWTP	SSOW#:		

Analysis Requested

Sample Identification	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA - PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:				N		
EFFLUENT	6-4-19	1315	G	Water	N	N	X	2	
EFFLUENT DUPLICATE	6-4-19	1317	G	Water	N	N	X	2	
EQUIPMENT BLANK	6-4-19	1312	G	Water	N	N	X	2	
FIELD BLANK	6-4-19	1308	G	Water	N	N	X	2	

190-19915 Chain of Custody

Preservation Codes:

A - HCL	M - Hexane
B - NaOH	N - None
C - Zn Acetate	O - AsNaO2
D - Nitric Acid	P - Na2O4S
E - NaHSO4	Q - Na2SO3
F - MeOH	R - Na2S2O3
G - Amchlor	S - H2SO4
H - Ascorbic Acid	T - TSP Dodecahydrate
I - Ice	U - Acetone
J - DI Water	V - MCAA
K - EDTA	W - pH 4-5
L - EDA	Z - other (specify)

Other:

Possible Hazard Identification
☐ Non-Hazard ☐ Flammable ☐ Skin Irritant ☐ Poison B ☐ Unknown ☐ Radiological

Deliverable Requested: I, II, III, IV, Other (specify)

Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)
☐ Return To Client ☒ Disposal By Lab ☐ Archive For ______ Months

Special Instructions/QC Requirements:

Empty Kit Relinquished by:		Date:	Time:		Method of Shipment: UPS Ground	
Relinquished by: CA Tomaszewski	Date/Time: 6-4-2019/1400	Company: City of Cadillac	Received by: Angela M...		Date/Time: 6/5/19 12:15	Company: ETA MI
Relinquished by:	Date/Time:	Company	Received by:		Date/Time:	Company
Relinquished by:	Date/Time:	Company	Received by:		Date/Time:	Company

Custody Seals Intact: Δ Yes Δ No	Custody Seal No.:	Cooler Temperature(s) °C and Other Remarks:

Ver: 01/16/2019

TestAmerica

Cooler/Sample Receipt

(AFTER HOURS receipt- complete gray areas. Place cooler in walk-in, place this form in Receiving in-box. Date/Time rec'd ______ Initials ______)

☐ MSDS or Known Hazard Information Supplied by Client
☐ Bottle stickers applied ☐ ELEMENT comment entered ☐ MSDS COC scanned emailed to EH&S
☐ Discrepancies
☐ Short Hold
☐ Rush ☐24hr ☐2day ☐3day ☐5day ☐other

Client ID City of Cadillac
Work Order # 190-19915
Receipt evaluation performed by - Initials AMY Date 6/5/19 Time 12:15

Method of Shipment:
☐ Walk-In Client ☐ TestAmerica Field/Courier
☐ Other Client/3rd Party Courier ______
☐ Fed Ex Tracking # ______
☒ UPS Tracking # 1Z E47 491 03 5595 0229
☐ Other Ground

Shipping Container Type:
☒ Cooler ☐ Box
☐ None ☐ Other ______

Packing Materials:
☒ Plastic Bags ☐ Foam
☒ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☐ Other ______

Custody Seals Intact:
☒ Yes 690880/1 ☐ No
☐ N/A (not used or required)

Cooling Materials:
☒ Ice (solid) ☒ Ice (Melted)
☐ Blue Ice ☐ None
☐ Other ______

Bacteriological Samples	Temp (°C) Corrected	Frozen yes no	Received within 2 hours yes no	Sample Flagged yes no

Receipt Temperatures C.F. ☐ Check if Additional Sheets Required

C.F.	Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Received on same day sampled?	Acceptable?*	Cooler ID	Note Affected Samples if temperature not acceptable
	140252483			☐	☐	☐Yes ☐No	☐Yes ☐No		
	140252476			☐	☐	☐Yes ☐No	☐Yes ☐No		
-0.1	CP313207	4.1	3.9	☐	☒	☐Yes ☒No	☒Yes ☐No		

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 *Sample Receipt and Login, Attachment 2 – Holding Times, Preservation and Container Requirements*

Receipt Questions**	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	✓			
Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	✓			
Appropriate containers used & adequate volume provided?	✓			Preserved Bottles Checked with pH Strips* Yes No
Number of sample containers match COC?	✓			
Samples received within hold time?	✓			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			✓	
Was a Trip Blank received with VOA samples?			✓	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	✓			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	✓			

** May not be applicable if samples are not for compliance testing

* Excludes FOG, Volatiles, TOC Vials

Client Contact Record

Contact via: ☐ Phone ☐ Email ☐ Other ______ Person Contacted: ______ Date/Time: ______
☐ Discrepancy allowance agreement is on record in the client project file

Discussion/Resolution:

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory.

Reviewed by PM Signature [signature] Date 6/5/19

WI Page 1 of 1

WI No. DT-SCA-WI-001 10 effective 06/11/12